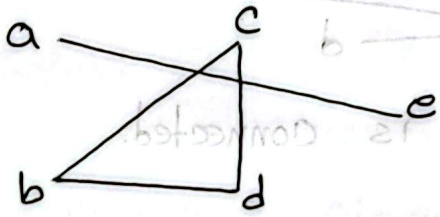
Fall 24

1)a

1) For,

$E1 = \{ae, bc, bd, cd\}$;

Graph -



So the graph is not connected.

Degrees -

a = 1 d = 2

b = 2 e = 1

c = 2

As degrees are not equal. Its not regular.

As the graph is disconnected, it doesn't have Eular trial.

Also there is no Eulerian circuit and Hamiltonian circuit.

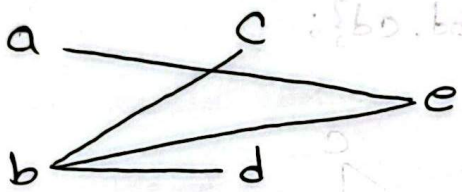
The graph is also not a tree.



ii) For,

$E2 = \{ae, bc, bd, be\}$

Graph.



So the graph is connected.

Degrees-

$a = 1$ $c = 1$ $e = 2$

$b = 3$ $d = 1$

So, not regular (Unequal degrees)

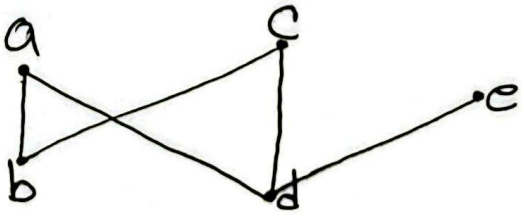
There is no Eulerian trial, Eulerian circuit and Hamiltonian circuit.

The graph is a tree.

iii) For,

$E3 = \{a;b, ad, bc, cd, de\}$

Graph-



The graph is connected.

Degrees -

$a = 2$ $c = 2$ $e = 1$

$b = 2$ $d = 2$

So, the graph is not regular.

There is an Eulerian trial (e-d-a-b-c-d)

There is no Eulerian circuit.

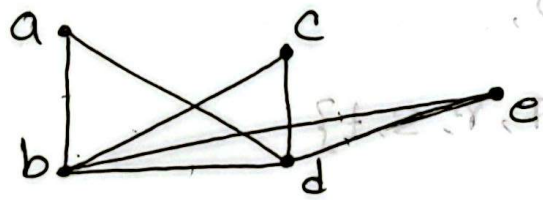
There is no Hamiltonial circuit

The graph is not a tree.

iv) For,

$E4 = \{ab, ad, bc, bd, be, cd, de\}$

Graph-



The graph is connected.

Degrees- $a = 2$ $c = 2$ $e = 2$

$b = 4$ $d = 4$

So, the graph is not regular.

There is an Eulerian trial. (a,b,d,e,b,c,d,a)

There is an Eulerian circuit. (a,b,d,e,b,c,d,a)

There is no Hamiltoniah circuit.

The graph is not a tree.

b) ✱

For (V, E1),

$V = \{a, b, c, d, e\}$

$E1 = \{ab, ad, bc, bd, be, cd, de\}$

Vertex	Degree
a	2
b	4
c	2
d	4
e	2

Degree Sequence -

(4, 4, 2, 2, 2)

For (V0, E0),



$V0 = \{P, q, r, s, t\}$

$E0 = \{Pq, Pr, Ps, Pt, qr, qs, qt\}$

Vertex	Degree
P	4
q	4
r	2
s	2
t	2

Degree sequence -

(4, 4, 2, 2, 2)

Vertices matching -

For degree 4,

b = p

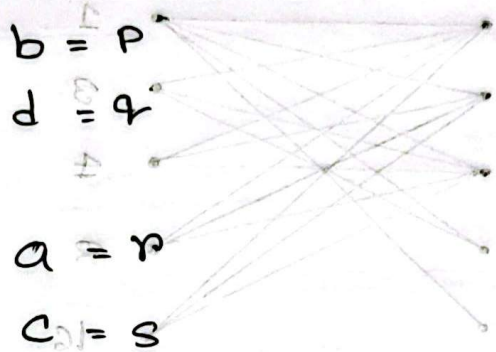
d = q

For degree 2,

a = r

c = s

e = t



Now,

$|V| = |V0|$, $|E1| = |E0|$

Number of vertices with degree n is same for both graph.

So the graphs are isomorphic.

2.b

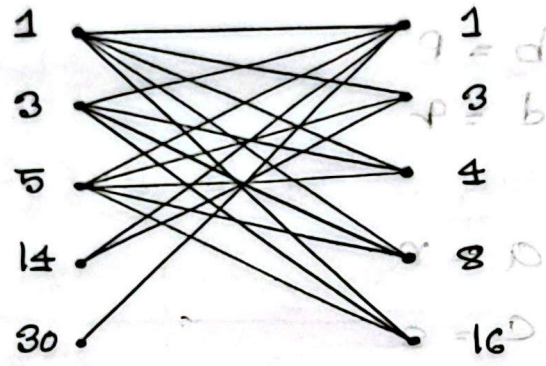
Given,

$X = \{1, 3, 5, 14, 30\}$

$Y = \{1, 3, 4, 8, 16\}$

GCD 1 ?	y=1	y=3	y=4	y=8	y=16
x=1	1	1	1	1	1
x=3	1	0	1	1	1
x=5	1	1	1	1	1
x=14	1	1	0	0	0
x=30	1	0	0	0	0

So, Bipartite graph -



∴ Adjacency matrix,

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Q-3.a

a) a

b) a, b, c, d, f, h, j, q, t.

c) e, g, i, k, l, m, n, o, p, r, s, u.

d) q, r.

e) a

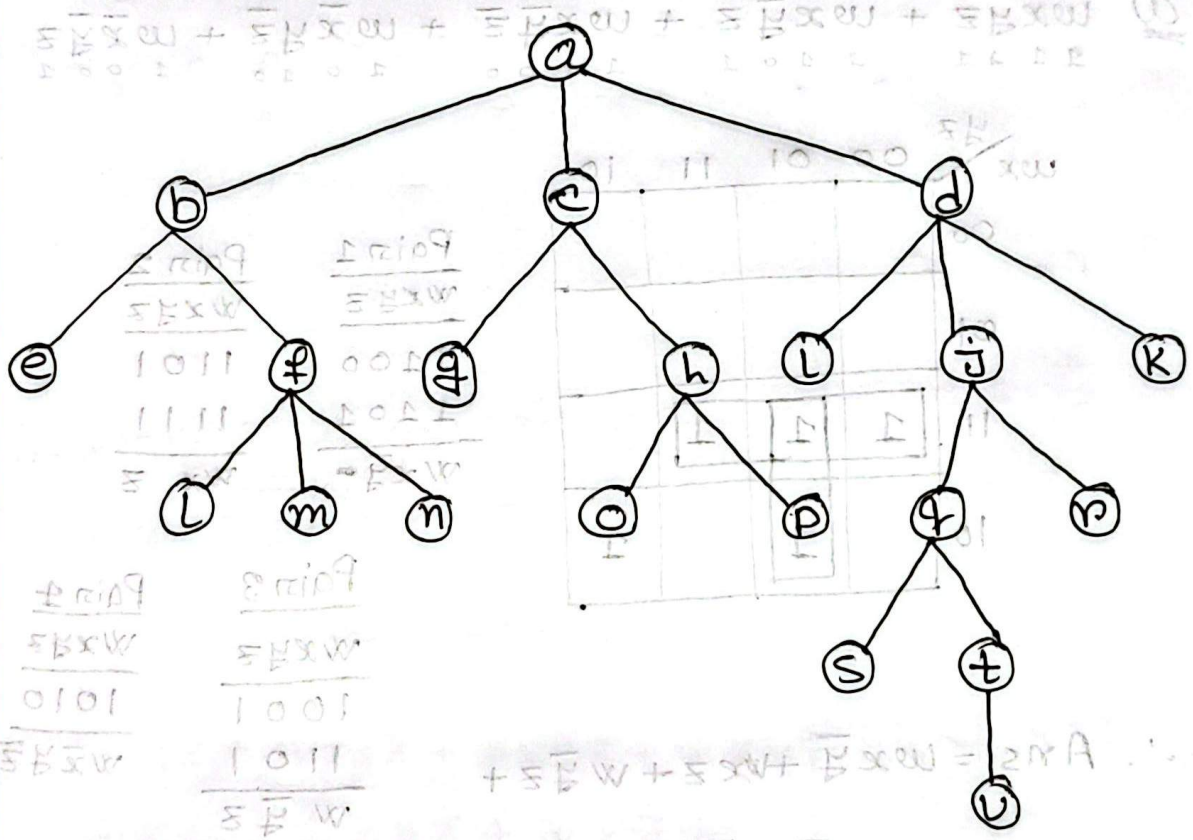
f) p

g) f, b, a

h) e, f, l, m, n.

b)

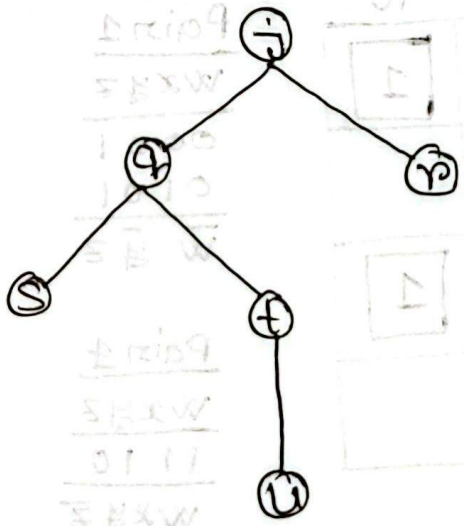
i) For a,



ii) For e,

There is no subtree for e. As e is a leaf.

iii) For j,



4. a

a) $wxyz + w\bar{x}yz + w\bar{x}\bar{y}z + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}\bar{y}z$

wx \ yz	00	01	11	10
00				
01				
11	1	1	1	
10		1		1

Pair 1

w	x	y	z
1	1	0	0
1	1	0	1
1	1	-	-

$wx\bar{y}$

Pair 2

w	x	y	z
1	1	0	1
1	1	1	1
1	1	-	-

wx

Pair 3

w	x	y	z
1	0	0	1
1	0	1	1
1	0	-	1

$w\bar{y}z$

Pair 4

w	x	y	z
0	1	0	1
1	0	1	0
-	1	-	-

$\bar{w}xz$

$\therefore Ans = wx\bar{y} + wx + w\bar{y}z + \bar{w}xz$

b) $\bar{w}\bar{x}\bar{y}z + \bar{w}\bar{x}y\bar{z} + \bar{w}xy\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}z + w\bar{x}yz + wxyz$

wx \ yz	00	01	11	10
00		1		1
01		1		
11		1		1
10			1	

Pair 1

w	x	y	z
0	0	0	1
1	0	0	0
-	0	0	-

$\bar{x}\bar{y}$

Pair 2

w	x	y	z
0	1	0	1
1	1	0	1
-	1	0	-

$x\bar{y}z$

Pair 3

w	x	y	z
0	0	1	0
-	-	-	-

$\bar{w}\bar{x}y\bar{z}$

Pair 4

w	x	y	z
1	1	1	0
-	-	-	-

$wxy\bar{z}$

Pair 5

w	x	y	z
1	1	0	1
-	-	-	-

$wx\bar{y}z$

$\therefore Ans = wx\bar{y}z + wxy\bar{z} + x\bar{y}z + \bar{x}\bar{y} + \bar{w}\bar{x}y\bar{z}$

c) $w\bar{x}\bar{y}z + w\bar{x}y\bar{z} + wx\bar{y}z + wxy\bar{z} + \bar{w}\bar{x}\bar{y}z + \bar{w}\bar{x}y\bar{z} + \bar{w}xy\bar{z} + \bar{w}x\bar{y}z$

=>

wx \ yz	00	01	11	10
00		1		1
01		1		
11		1	1	1
10	1	1		

Pair 1

w	x	y	z
0	0	0	1
0	1	0	1
1	1	0	1
1	0	0	1
			$\bar{y}z$

Pair 2

w	x	y	z
1	0	0	0
1	0	0	1
			$w\bar{x}\bar{y}$

Pair 3

w	x	y	z
1	1	1	1
1	1	1	0
			wxy

Pair 4

w	x	y	z
0	1	0	0
			$\bar{w}x\bar{y}$

$\therefore Ans = \bar{y}z + w\bar{x}\bar{y} + \bar{w}x\bar{y} + \bar{x}\bar{y}z$

d) $\bar{w}\bar{x}\bar{y}z + \bar{w}\bar{x}y\bar{z} + \bar{w}xy\bar{z} + \bar{w}x\bar{y}z + w\bar{x}\bar{y}z + w\bar{x}y\bar{z} + wxy\bar{z} + wx\bar{y}z + wx\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z}$

=>

wx \ yz	00	01	11	10
00		1	1	1
01			1	
11		1	1	1
10			1	1

Pair 1

w	x	y	z
0	0	1	1
0	1	1	1
1	1	1	1
1	0	1	1
			$\bar{x}z$

Pair 2

w	x	y	z
1	1	1	1
1	1	1	0
1	0	1	1
1	0	1	0
			wy

Pair 3

w	x	y	z
0	0	1	1
0	1	0	1
1	0	1	1
1	0	1	0
			$\bar{x}y$

Pair 4

w	x	y	z
0	0	0	1
0	1	0	0
			$\bar{w}\bar{x}z$

Pair 5

w	x	y	z
1	1	1	1
1	0	1	1
			wxz

$\therefore \underline{Ans} = \bar{x}z + wy + \bar{w}\bar{x}z + \bar{w}xy$

5.a

a) $M_a = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ $\therefore M_a^T = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

As, Diagonal of $M_a = 1, 0, 1, 1$

This matrix is not Reflexive. (Requires 1)

This matrix is not Irreflexive. (Requires 0)

As, $M_a = M_a^T$, it is symmetric.

Now, Let,

$M_a = (a,a), (a,b), (a,d), (b,a), (b,c), (c,b), (c,c), (c,d), (d,a), (d,c), (d,d)$.

So, its not antisymmetric.

Also, its not transitive.

b)

$$M_b = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \therefore M_b^T = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

As Diagonal of $M_b = (1, 1, 1, 1)$.

Its a Reflexive Matrix.

Its not an Irreflexive Matrix.

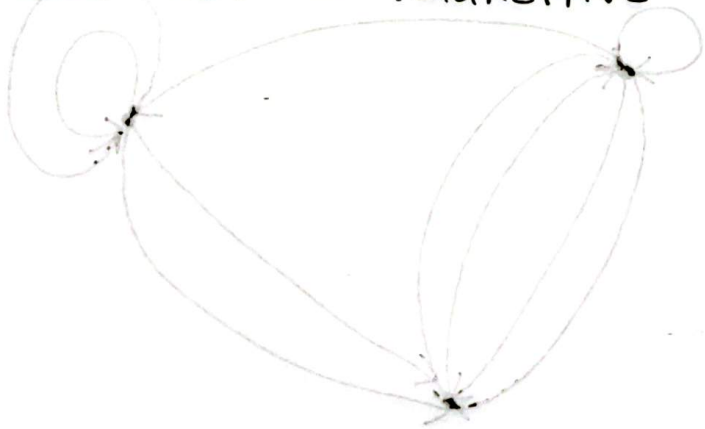
As $M_b \neq M_b^T$, Its not symmetric.

Now let,

$M_b = (a,a), (a,b), (a,c), (b,b), (c,c), (c,d), (d,a), (d,d)$

So, its an antisymmetric matrix.

Its also not a Transitive Matrix.



Spring 25

2. a) (OR)

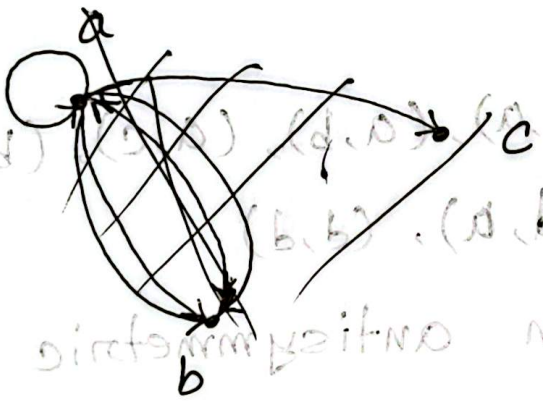
1) For,

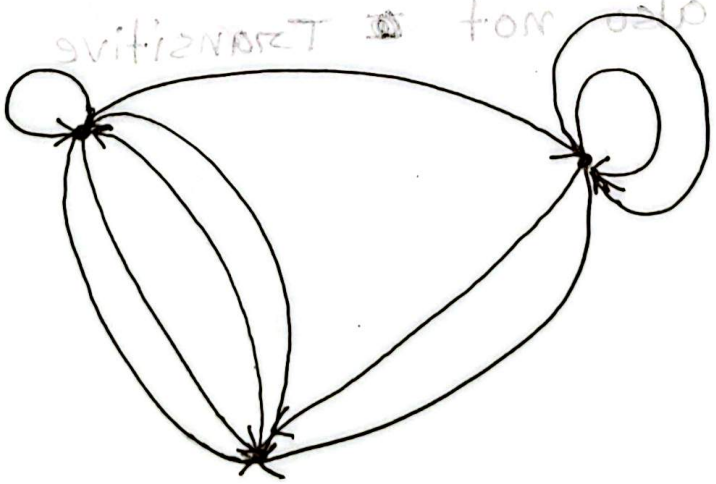
	a	b	c
a	1	2	1
b	2	0	0
c	0	2	2

let,

M_A = (a,a), 2(a,b), 2(a,c), 2(b,a), 2(c,b), 2(c,c).

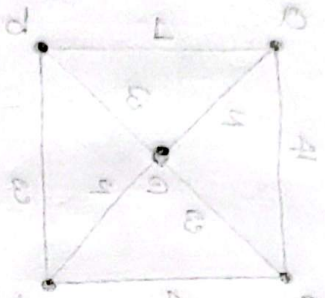
Graph –



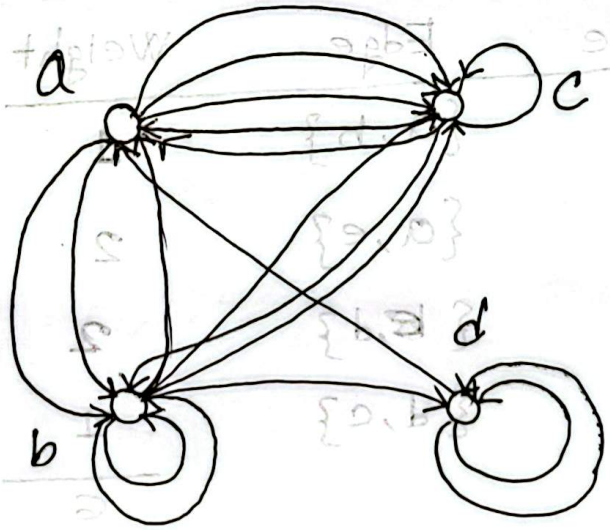


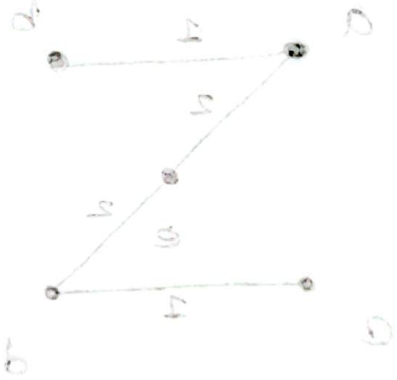
11) - For

	a	b	c	d
a	0	2	3	0
b	1	2	2	1
c	2	1	1	0
d	1	0	0	2



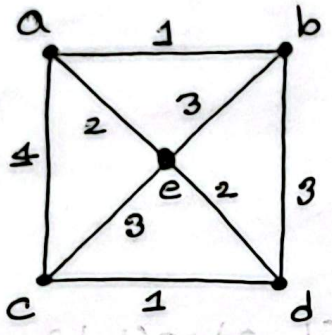
let, $m_B = 2(a,b), 3(a,c), (b,a), 2(b,b), 2(b,c), (b,d), 2(c,a), (c,b), (c,c), (d,a), 2(d,d)$





~~b)~~

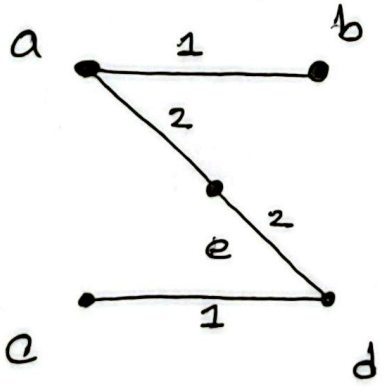
1)



Using Prim's algorithm –

Choice	Edge	Weight
1	{a, b}	1
2	{a, e}	2
3	{e, d}	2
4	{d, c}	1
		6

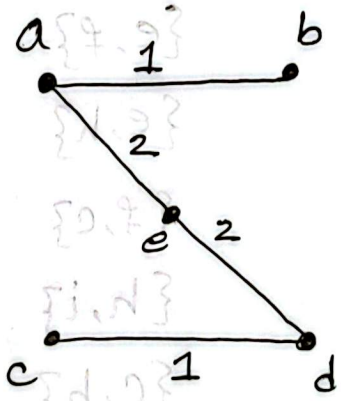
MST



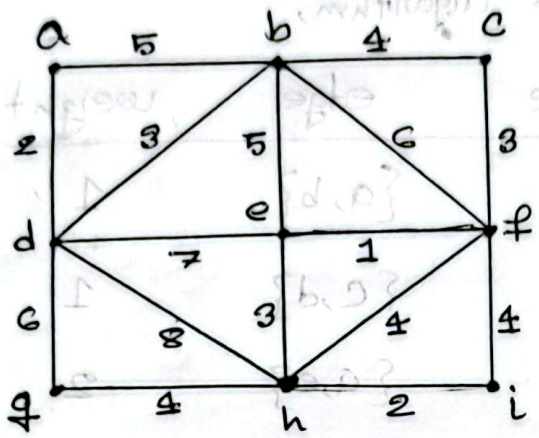
Using Kruskal's algorithm,

Choice	edge	weight
1	{a, b}	1
2	{c, d}	1
3	{a, e}	2
4	{e, d}	2
	Total-	6

MST -



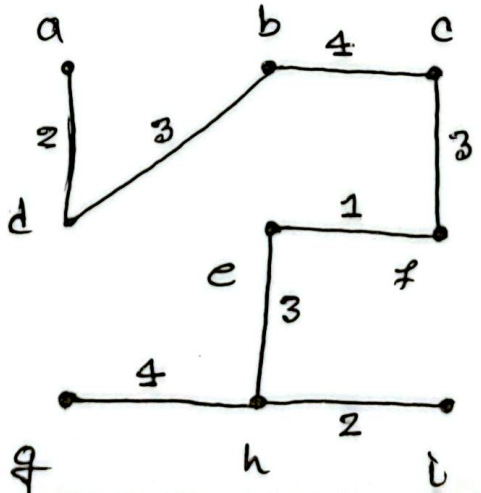
11)



Using Prims algo,

Choice	edge	weight
1	{e, f}	1
2	{e, h}	3
3	{f, c}	3
4	{h, i}	2
5	{c, b}	4
6	{b, d}	3
7	{d, a}	2
8	{h, g}	4
		Total :- 22

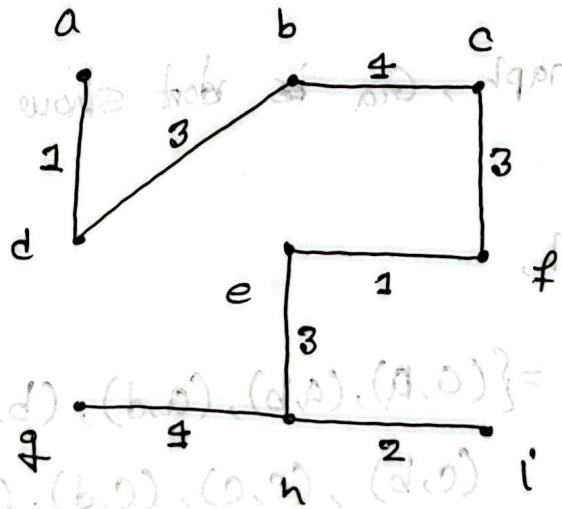
So, MST-



Using kruskal's algo-

Choice	edge	weight
1	{e, f}	1
2	{a, d}	2
3	{h, i}	2
4	{b, d}	3
5	{c, f}	3
6	{e, h}	3
7	{b, c}	4
8	{g, h}	4
		Total = 22

MST-



4.b

a) From graph,

$G_a = \{(a,a), (a,c), (a,d), (b,b), (c,a), (c,c), (d,a), (d,d)\}$

Reflexive

Yes, its a reflexive graph.

Symmetric

Yes, its a symmetric graph.

Transitive

$(c,d) \notin G_a$. So its not transitive.

So the graph, G_a dont show equivalence relation.

b) From graph,

$G_b = \{(a,a), (a,b), (a,d), (b,b), (b,a), (b,c), (c,b), (c,c), (c,d), (d,a), (d,c), (d,d)\}$

Reflexive

Yes, its a reflexive graph.

Symmetric

Yes, its a symmetric graph.

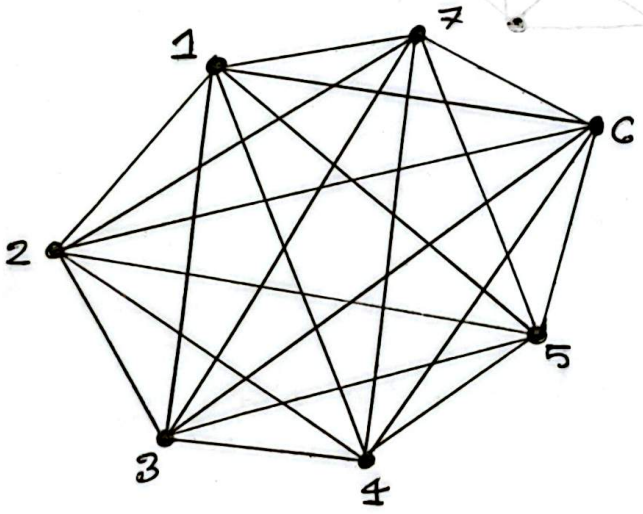
Transitive

$(a,c) \notin G_b$. So its not transitive.

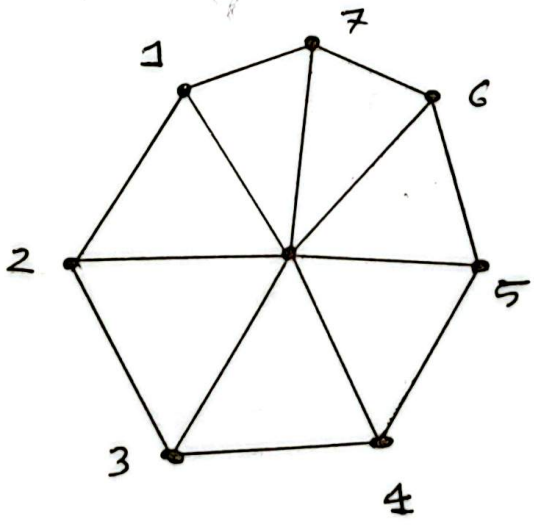
So the graph, G_b dont show equivalence relation.

5.a

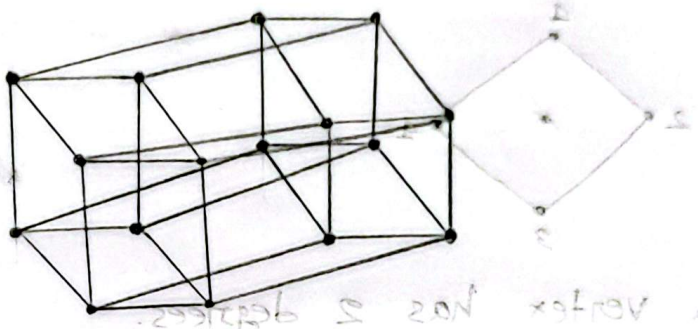
i) K_7 (Complete graph) -



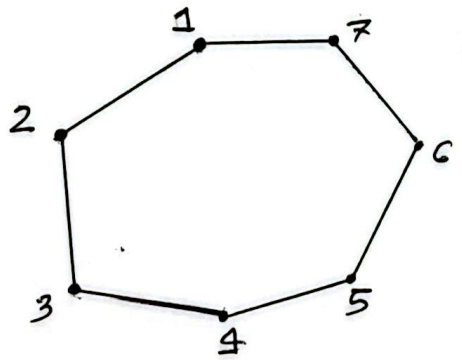
ii) W_7 (Wheel graph) -



III) Q_4 (Hypercube)-

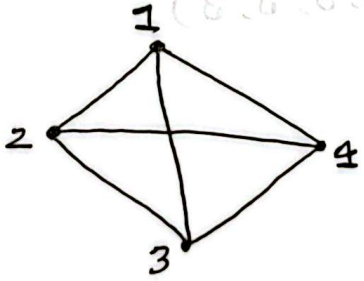


IV) C_7 (Cycle) -



b)

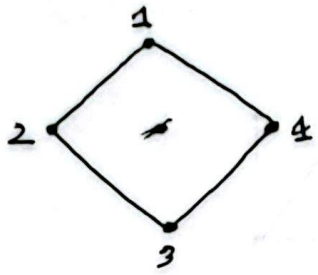
i) K_4



Eeach vertex has 3 degrees.

So sequence = (3, 3, 3, 3)

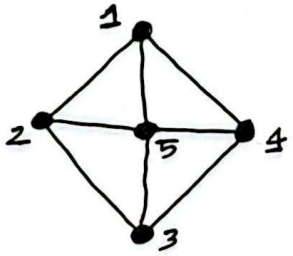
ii) C_4



Each vertex has 2 degrees.

So, sequence = (2, 2, 2, 2)

iii) W_4

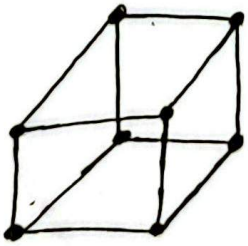


Each vertex has (1, 2, 3, 4) has 3 degrees.

5th vertex has 4 degrees.

So, sequence = (4, 3, 3, 3, 3)

iv) Q_3



Vertics = $2^3 = 8$.

Each vertex has 3 degrees.

So, sequence = (3, 3, 3, 3, 3, 3, 3, 3)